

Flower Classification Project Report

Prepared by: Arpita Panda

Abstract

This project focuses on classifying flower images into different categories using machine learning and image processing techniques. The system is trained on a flower dataset and predicts the flower class from an input image.

Objectives

1. Build an image classification model.
2. Classify flowers accurately.
3. Apply machine learning concepts to real-world image data.

Tools and Technologies

Python, Jupyter Notebook, NumPy, Pandas, Matplotlib, TensorFlow/Keras, Scikit-learn.

Methodology

Data collection, preprocessing, model training, evaluation, and prediction.

Results

The model successfully classifies flower images and demonstrates the practical use of machine learning in computer vision tasks.

Conclusion

Flower classification is an effective application of machine learning. The project improves understanding of image processing, datasets, and classification algorithms.

Future Scope

Improve accuracy with larger datasets, advanced neural networks, and deployment as a web or mobile application.